

RAVI KUMAR DUSI

Electronics Engineer

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☎ +49 17620555698

📍 Goslar, Germany

🌐 Ravi Dusi



WORK EXPERIENCE

Werkstudent Electronics & EMC Engineer

Denpaflux GmbH

📅 Sept 2023 – Sept 2024

📍 Munich

- Designed & simulated PCBs (analog/digital circuits, DC-DC converters, filters) using Altium Designer, Implementing EMC rules and layout optimization.
- Developed DC-DC buck converter and Sallen-Key filter PCBs, fabricated prototypes, and correlated results between Theory, Simulation, and Lab measurements.
- Performed Signal Integrity / Power Integrity analysis, controlled impedance routing, and applied IPC/IEC guidelines for high-speed interfaces in my Masters Thesis.
- Managed CAD component library (symbols, footprints, BOM library) and generated ODB++ and manufacturing documentation for EMC report & Production.
- Hands on experience handling electronics lab instruments such as oscilloscopes, multimeter, and vector network analyzers, soldering, Testing.
- Master's Thesis: Best practices in Electronics Design, High-Speed PCB Design Rules (focused on SI/PI techniques, EMC resilience, and robust design workflows).

Werkstudent Electronics Engineer

Konux GmbH

📅 Jan 2023 – July 2023

📍 Munich

- Successfully Designed the multi-layer and test bench system consists embedded electronics from analyze requirements, design circuits/controls until manufacturing, and assisted in python scripting.
- Validated microcontroller-based PCBs for railway monitoring systems, component selection, and circuit dimensioning.
- Maintained CAD library (footprints, parametric data alignment) to support design-to-manufacture consistency.
- Testing of the developed hardware units in the laboratory by Performing soldering, modifying measurement box and cables, Optimizing Hardware, PCB Assembly, circuit prototyping, debugging, Supported validation with lab equipment.

Intern Hardware Development

Rocket Factory Augsburg

📅 Sept 2022 – Dec 2022

📍 Augsburg

- Solid understanding of power electronic circuitry by Reviewing schematics of Power Distribution Unit(PDU).
- Testing of the developed hardware units (PDU) in the laboratory and also simulate it in system.

SUMMARY

Electronics engineer with a Master's degree in Micro-systems Electronics and hands-on expertise in PCB design, SI/PI, EMC/ESD testing, hardware validation, CAD library maintenance, and production documentation. Passionate about Hardware Development, High speed Design practices and ensuring robust, manufacturable solutions in Embedded and system-level architectures. Fast learner, detail-oriented, and highly motivated to contribute to Engineering Innovation.

SKILLS

- **Hardware Development:** Analog circuits, Signal Integrity, Power Integrity, Power Electronics, DC-DC Converters, Low Power Design, Semiconductor knowledge, Grounding techniques, EMC fundamentals, Filters.
- **PCB-ECAD:** Circuit Schematics, multilayer PCB layout, CAD library maintenance, Altium Designer, KiCAD, EMC-aware layout.
- **Simulation & Programming:** LTSpice simulation, Power & Noise Management, SI/PI analysis, Basic Python & Matlab for data analysis.
- **Hardware Testing & Validation:** Soldering (THT /SMD), EMC testing, Lab Equipments such as Oscilloscope, Multimeters & VNA, Analyzer, Validation, Failure analysis.
- **Documentation:** Technical documentation, ODB++ data preparation, manufacturing documentation (DFM), BOM creation, Assembly guidelines Test report (DFT), Prototyping & Debugging.
- **Communication Protocols:** USB, UART, SPI, I2C, High Speed Serial Interfaces, SerDes Knowledge.
- **Compliance & Standards:** EMC/ESD compliance, IEC 61000-4, IPC 2141A, IPC 2221.

EDUCATION

Masters in Electronics Engineering: Micro-systems

Hochschule Bremen

📅 (Sept 2021 – Sept 2024)

Thesis : Best practices in Electronics Design. Unveiling myths of common Practices.

Subjects: **Material Science, Micro-technology & Micro-structuring Analog & Mixed Circuit Design, Concept Engineering of Mixed Technology.**

Bachelors in Electronics & Communication Engineering

Sathyabama University

📅 (Jun 2016 – Mar 2020)

Thesis : Design and Development of Unmanned Aerial vehicle for plant disease detection and diagnosis.

Subjects: Electronic circuits, Electrical network, Control systems, C/C++ programming, Communication systems, Microprocessor, Digital Signal Processing.

Personal skills

Interdisciplinary team communication

Analytical thinking & Quick learner

Problem solving & Presentation skills

Hands on Lab experience

LANGUAGES

English

German



ADDITIONAL

- Available for the relocation & start Immediately.
- Knowledge and great interest in High speed PCB Design .
- Excited to work in international environment & gain further knowledge in Embedded & Hardware Development.

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September 30, 2025
Sinelec Technologies Deutschland
GmbH

SINELEC DEPARTMENT
BRACHVOGELWEG 9, 85375, NEUFABRN
BEI FREISING

RE : Unslociated Application for the Position in "Sinelec Technologies"

Dear Sinelec-Tech Team,

I am writing to express my keen interest in exploring opportunities at Sinelec-Techlogies. Your mission to deliver High-quality Engineering services, especially in areas like **Embedded systems, Electronic design, Hardware validation, and Simulation, aligns deeply with my professional passions and technical strengths.**

As a person who has Masters in Micro-systems Engineering and good knowledge of Electronics and Semiconductor working and manufacturing, along with Working experience in companies focused on **PCB design, signal & power integrity, EMC/ESD testing, CAD library management, and design-to-production documentation,** I have cultivated skills that match your service offerings. I thrive in environments where theory, simulation, and measurement come together to produce reliable, production-ready hardware. **The breadth of your engineering services suggests a collaborative, innovation-driven workplace, which is exactly where I see myself contributing, learning, and growing.**

I would welcome the opportunity to discuss how my background in PCB design, system integration, and prototyping can contribute to Sinelec-Tech's engineering teams. **I attach my CV for your consideration and would be honored to talk further.** I am ready to relocate & start work immediately also would expect a salary range of 55- 60 K/annum - Negotiable.

Thank you very much for your time and consideration. I look forward to the possibility of working with you.

Sincerely,

RAVI KUMAR Dusi

